Dr. Junsoo Lee

Last update: October 30, 2026

Contact Information

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Professional Positions

Assistant Professor 2022 – Present
Department of Mechanical Engineering, University of South Carolina

Graduate Research Assistant 2019 – 2022
School of Aerospace Engineering, Georgia Institute of Technology

Graduate Teaching Assistant 2018 – 2019
School of Aerospace Engineering, Georgia Institute of Technology

Graduate Research Assistant 2016 – 2018
Department of Aerospace Engineering, Seoul National University

Education

Ph.D. in Aerospace Engineering
Georgia Institute of Technology, 2022

M.S. in Mathematics
Georgia Institute of Technology, 2021

M.S. in Aerospace Engineering
Seoul National University, 2018

B.S. in Mechanical and Aerospace Engineering
Seoul National University, 2016

Honors & Awards

Nominee, Distinguished Undergraduate Research Mentor Award 2026
2027 award, Office of Undergraduate Research, University of South Carolina

Nominee, Distinguished Undergraduate Research Mentor Award 2025
2026 award, Office of Undergraduate Research, University of South Carolina

Research and Education Awards Program (REAP) Award 2024
NASA South Carolina Space Grant Consortium

IMECE Faculty Travel Grant 2024
National Science Foundation

Teaching Innovation Award 2023
Department of Mechanical Engineering, University of South Carolina, for EMCH 367

Faculty Travel Grant 2023
Korean-American Scientists and Engineers Association (KSEA)

Faces of Inclusive Excellence 2021
Institute of Diversity, Equity, and Inclusion, Georgia Institute of Technology

Research Support

Completed

Implicit Discretization for Finite-Time and Fixed-Time Control 2024 – 2025
Source of Support: NASA South Carolina Space Grant Consortium, REAP
Role: PI

Pending

Launch, Station-Keeping, and Recovery of Shipboard Tethered Multirotor UAVs Through Cooperative Winch–Vehicle Control 2027 – 2030
Source of Support: Office of Naval Research, Young Investigator Program
Role: PI

CAREER: Capability-Aware Autonomy for Retractable-Tether Aerial Robots in Confined Environments 2027 – 2031
Source of Support: National Science Foundation
Role: PI, $599,921

A Dual-Process Learning-Unlearning Architecture for Long-Term Autonomy under Progressive Battery Degradation 2027 – 2030
Source of Support: National Science Foundation
Role: PI, $403,484

Reducing Recurrent Flow-Related Patient-Ventilator Asynchrony with Safe Patient-Specific Adaptive Control 2027 – 2028
Source of Support: SC EPSCoR
Role: PI, $60,000

Publications

Students supervised by the PI are underlined.

Archival Journal Publications:

J1. K. Bojappa and **J. Lee**, "Review on Particle Swarm Optimization: Application Toward Autonomous Dynamical Systems," *IEEE/CAA Journal of Automatica Sinica*, vol. 12, no. 9, pp. 1762-1775, 2025.

J2. D. Handrick, M. Eckenrode, and **J. Lee**, "Review of Tethered Unmanned Aerial Vehicles: Building Versatile and Robust Tethered Multirotor UAV

System," *Dynamics*, vol. 5, no. 2, article no. 17, 2025.

J3. **J. Lee**, S. P. Bhat, and W. M. Haddad, "Nontangency-Based Lyapunov Tests for Convergence and Stability in Discrete-Time Systems," *SIAM Journal on Applied Dynamical Systems*, vol. 24, no. 2, pp. 1711-1751, 2025.

J4. J. Park, S. Lee, and **J. Lee**, "Motion Planning of Quadrotor UAV via Reinforcement Learning in Unknown Obstacle Environment," *International Journal of Control, Automation and Systems*, vol. 23, no. 10, pp. 3035-3047, 2025.

J5. **J. Lee**, W. M. Haddad, and M. Lanchares, "Correction to "Finite Time Stability and Optimal Finite Time Stabilization for Discrete-Time Stochastic Dynamical Systems"," *IEEE Transactions on Automatic Control*, vol. 70, no. 8, pp. 5624-5626, 2025.

J6. **J. Lee** and W. M. Haddad, "Fixed Time Stability of Discrete-Time Stochastic Dynamical Systems," *Automatica*, vol. 163, article no. 111553, 2024.

J7. **J. Lee**, "Geometric Semistability for Ultimately Bounded Semistability of Discrete-Time Dynamical Systems," *Systems & Control Letters*, vol. 188, article no. 105787, 2024.

J8. **J. Lee**, W. M. Haddad, and M. Lanchares, "Finite Time Stability and Optimal Finite Time Stabilization for Discrete-Time Stochastic Dynamical Systems," *IEEE Transactions on Automatic Control*, vol. 68, no. 7, pp. 3978-3991, 2023.

J9. **J. Lee** and W. M. Haddad, "Fixed Time Stability and Optimal Stabilisation of Discrete Autonomous Systems," *International Journal of Control*, vol. 96, no. 9, pp. 2341-2355, 2023.

J10. W. M. Haddad and **J. Lee**, "Finite-Time Stabilization and Optimal Feedback Control for Nonlinear Discrete-Time Systems," *IEEE Transactions on Automatic Control*, vol. 68, no. 3, pp. 1685-1691, 2023.

J11. W. M. Haddad, **J. Lee**, and S. P. Bhat, "Asymptotic and Finite Time Semistability for Nonlinear Discrete-Time Systems with Application to Network Consensus," *IEEE Transactions on Automatic Control*, vol. 68, no. 2, pp. 766-781, 2023.

J12. W. M. Haddad and **J. Lee**, "Lyapunov Theorems for Stability and Semistability of Discrete-Time Stochastic Systems," *Automatica*, vol. 142, article no. 110393, 2022.

J13. **J. Lee** and W. M. Haddad, "On Finite-Time Stability and Stabilization of Nonlinear Hybrid Dynamical Systems," *AIMS Mathematics*, vol. 6, no. 6, pp. 5535-5562, 2021.

J14. W. M. Haddad and **J. Lee**, "Finite-Time Stability of Discrete Autonomous Systems," *Automatica*, vol. 122, article no. 109282, 2020.

Journal Articles under Review and in Preparation:

R1. **J. Lee**, "Closed-Form Consistent Discretization of Exponential Predefined-Time Controllers," *IEEE Control Systems Letters* with the American Control Conference 2027 option, under review, 2026.

R2. S. Jadhav and **J. Lee**, "Review on Dual Process Theory: Long-term Autonomous Mission," in preparation for *IEEE Transactions on Cognitive and Developmental Systems*.

Conference Proceedings:

C1. S. Jadhav, D. Walters, A. Keller, and **J. Lee**, "Long-Term Autonomous Mission Planning Under Battery Degradation," in proceedings of *AIAA SciTech Forum*, Orlando, FL, January 2027, to appear.

C2. D. Handrick, K. Bojappa, H. Sharifi, **J. Lee**, A. Downey, and T. Crawford, "NV ODMR Magnetometer for Geomagnetically Induced Current Estimation Toward Electric Grid Resilience," in proceedings of *AIAA SciTech Forum*, Orlando, FL, January 2027, to appear.

C3. H. Sharifi, A. Varillas, D. Handrick, P. Bailey, A. Diaz, J. Giannasi, T. Crawford, A. Downey, **J. Lee**, and W. De Backer, "Investigation of Microwave Field Effects on the ODMR Response of NV Centers in Diamond Using a Coplanar Waveguide," in proceedings of *AIAA SciTech Forum*, Orlando, FL, January 2027, to appear.

C4. **J. Lee**, "Arc-Length-Based Convergence and Semistability Tests for Discrete-Time Systems," in proceedings of *IEEE Conference on Decision and Control*, Honolulu, HI, December 2026, to appear.

C5. E. Sill, K. Bojappa, and **J. Lee**, "Thermodynamic Particle Swarm Optimization for Multi-Agent Target Search in Unknown Obstacle Environments," in proceedings of *ASME International Mechanical Engineering Congress and Exposition*, Vancouver, BC, Canada, November 2026, to appear.

C6. **J. Lee** and H. Lutz, "Implicit Discretization for Fixed-Time Stabilization of Discrete-Time Nonlinear Systems," in proceedings of *Modeling, Estimation and Control Conference*, Phoenix, AZ, 2026.

C7. **J. Lee** and H. Lutz, "Consistent Finite-Time Stabilization and Optimal Control for Nonlinear Discrete-Time Systems," in proceedings of *Modeling, Estimation and Control Conference*, Phoenix, AZ, 2026.

C8. K. Bojappa and **J. Lee**, "Thermodynamic Particle Swarm Optimization for Multi-agent System in Unknown Environment," *IFAC-PapersOnLine* (Modeling, Estimation and Control Conference), vol. 59, no. 30, pp. 461-466, 2025.

C9. **J. Lee**, "Lyapunov-like Theorems for Ultimately Bounded Discrete-Time Stochastic Systems," *IFAC-PapersOnLine* (Modeling, Estimation and Control Conference), vol. 59, no. 30, pp. 881-886, 2025.

C10. **J. Lee**, S. P. Bhat, and W. M. Haddad, "Nontangency-Based Stability Tests for Discrete-Time Dynamical Systems," in proceedings of *IEEE American Control Conference*, pp. 2368-2373, Denver, CO, 2025.

C11. **J. Lee** and W. M. Haddad, "Fixed Time Stability of Discrete-Time Stochastic Dynamical Systems," in proceedings of *IEEE American Control Conference*, pp. 4001-4006, San Diego, CA, 2023.

C12. **J. Lee**, S. P. Bhat, and W. M. Haddad, "Nontangency-Based Lyapunov Tests for Convergence in Discrete-Time Dynamical Systems," in proceedings of *IEEE American Control Conference*, pp. 2424-2429, San Diego, CA, 2023.

C13. **J. Lee** and W. M. Haddad, "Lyapunov Theorems for Finite Time and Fixed Time Semistability of Discrete-Time Stochastic Systems," in proceedings of *IEEE Aerospace Conference*, pp. 1-10, Big Sky, MT, 2023.

C14. **J. Lee** and W. M. Haddad, "Finite and Fixed Time Semistability and Consensus for Discrete-Time Network Systems," in proceedings of *IEEE Mediterranean Conference on Control and Automation*, pp. 809-814, Athens, Greece, 2022.

C15. **J. Lee** and W. M. Haddad, "Fixed Time Stability of Discrete Autonomous Systems," in proceedings of *IEEE Mediterranean Conference on Control and Automation*, pp. 526-531, Athens, Greece, 2022.

C16. **J. Lee**, W. M. Haddad, and M. Lanchares, "Optimal Finite Time Control for Discrete-Time Stochastic Dynamical Systems," in proceedings of *IEEE American Control Conference*, pp. 3500-3505, Atlanta, GA, 2022.

C17. **J. Lee**, W. M. Haddad, and S. P. Bhat, "Stochastic Finite Time Stability of Discrete-Time Systems," in proceedings of *IEEE Conference on Decision and Control*, pp. 6646-6651, Austin, TX, 2021.

C18. W. M. Haddad and **J. Lee**, "A Thermodynamic-Based Control Architecture for Semistability and Consensus of Discrete-Time Nonlinear Multiagent Systems," in proceedings of *IEEE Conference on Control Technology and Applications*, pp. 499-504, San Diego, CA, 2021.

C19. W. M. Haddad and **J. Lee**, "Finite-Time Stabilization and Optimal Feedback Control for Nonlinear Discrete-Time Systems," in proceedings of *IEEE Conference on Control Technology and Applications*, pp. 202-207, San Diego, CA, 2021.

C20. W. M. Haddad and **J. Lee**, "Lyapunov Theorems for Semistability of Discrete-Time Stochastic Systems with Application to Network Consensus with Random Communication Noise," in proceedings of *IEEE Mediterranean Conference on Control and Automation*, pp. 892-897, Bari, Italy, 2021.

C21. W. M. Haddad, **J. Lee**, and S. P. Bhat, "Asymptotic and Finite Time Semistability for Nonlinear Discrete-Time Systems," in proceedings of *IEEE Mediterranean Conference on Control and Automation*, pp. 1281-1286, Bari, Italy, 2021.

C22. W. M. Haddad and **J. Lee**, "Finite-Time Stability of Discrete Autonomous Systems," in proceedings of *American Control Conference*, pp. 5188-5193, Denver, CO, 2020.

C23. **J. Lee**, N. Cho, and Y. Kim, "Smooth Trajectory Generation and Control of Multirotor with Slung Payload," *Asia-Pacific International Symposium on Aerospace Technology*, Seoul, Republic of Korea, 2017.

C24. S. Jeong, K. Oh, Y. Kim, B. Han, **J. Lee**, S. Seo, and Y. Kim, "Precise Ephemeris Generation of the Korean Positioning System Using Particle Swarm Optimization," *Korean Society for Aeronautical and Space Sciences Conference*, pp. 376-377, 2017.

C25. **J. Lee**, S. Lee, S. H. Lee, and Y. Kim, "Modelling and Simulation of a 6-DOF Quadrotor," *Korean Society for Aeronautical and Space Sciences Conference*, pp. 204-205, Samcheok, Republic of Korea, April 2017.

C26. **J. Lee** and Y. Kim, "Fault Tolerant Controller of Quadrotor Propeller Failure Using Sliding Mode Control Scheme," *Korean Society for Aeronautical and Space Sciences Conference*, pp. 215-216, Jeju, Republic of Korea, November 2016.

Textbooks and Monographs:

B1. W. M. Haddad, Q. Hui, and **J. Lee**, *Network Information Systems: A Dynamical Systems Approach*, Philadelphia, PA: Society for Industrial and Applied Mathematics, 2023.

Book Chapters:

BC1. W. M. Haddad and **J. Lee**, "Lyapunov Analysis for Discrete-Time Stochastic Systems with Application to Network Consensus," in *Smarter Cyber Physical Systems: Enabling Methodologies and Applications*, K. Vamvoudakis and F. L. Lewis, Eds., Boca Raton, FL: CRC Press, 2025, p. 113.

Datasets and Software:

D1. **J. Lee**, "RTMUAV-Bench v0.1: Preliminary Simulation Environment and Envelope Datasets for Retractable-Tether Aerial Autonomy," Zenodo, 2026.

Poster Presentations:

P1. D. Walters, A. Keller, and **J. Lee**, "Battery-Aware Multi-Layer Decision Making Framework for Multirotor Unmanned Aerial Vehicle Long-Term Autonomous Mission," Summer Research Symposium, Columbia, SC, July 2026.

P2. H. Lutz and **J. Lee**, "Finite Time High Precision Control of Robotic Arm Manipulator," Discover USC, Columbia, SC, April 2026.

P3. S. Yadav, D. Handrick, and **J. Lee**, "Robust Control Framework for Retractable Tether Multirotor UAVs (RTMUAV)," Discover USC, Columbia, SC, April 2026.

P4. A. Spina and **J. Lee**, "Flywheel Mechanism-Based Human-Mounted Fixed-Wing Unmanned Aerial Vehicle Launch Platform," Discover USC, Columbia, SC, April 2026 (First Place, Engineering and Computing C session).

P5. H. Lutz and **J. Lee**, "Finite Time Control and Set Valued Map," Modeling, Estimation and Control Conference, Pittsburgh, PA, October 2025.

P6. S. Yadav, D. Handrick, and **J. Lee**, "Retractable Tether Multirotor Unmanned Aerial Vehicle," Modeling, Estimation and Control Conference, Pittsburgh, PA, October 2025.

P7. K. Bojappa and **J. Lee**, "Thermodynamic Particle Swarm Optimization for Multi-agent System in Unknown Environment," Discover USC, Columbia, SC, April 2025.

P8. H. Lutz and **J. Lee**, "Realizing Digital Finite Time Control by Implicit Discretization," Discover USC, Columbia, SC, April 2025.

P9. M. Eckenrode, D. Handrick, and **J. Lee**, "Collaborative Tethered Unmanned Aerial Vehicle," Discover USC, Columbia, SC, April 2025.

P10. K. Bojappa and **J. Lee**, "Particle Swarm Optimization for Rendezvous Point Selection for Multi-Agent Systems in an Unknown Environment," Discover USC, Columbia, SC, April 2024.

P11. D. Handrick and **J. Lee**, "Tethered Unmanned Aerial Vehicles: Cable Dynamics and Collaborative Controls," Discover USC, Columbia, SC, April 2024.

P12. **J. Lee**, "Fixed Time Semistability for Discrete-Time Network Systems," UKC, Washington DC, August 2022.

Invited Talks & Workshops

Invited Talks

T1. **J. Lee**, "Designing Autonomous Networked Systems," Universidad San Francisco de Quito, Online, February 2026.

T2. **J. Lee**, "Achieving Finite Time Stability," *Smart Air Mobility Seminar*, Korea Aerospace University, Online, July 2024.

T3. **J. Lee**, "Application of Dynamical System Theory to Stochastic Dynamical System," *Smart Air Mobility Seminar*, Korea Aerospace University, Online, July 2023.

T4. **J. Lee**, "Networked Control System: Dynamical Systems Approach," *University of South Carolina Research Seminar Series*, Columbia, SC, US, March 2023.

T5. **J. Lee**, "Research Outside of Boundary," *BC-UofSC exploreCSR 2022-2023*, Columbia, SC, US, February 2023.

T6. **J. Lee**, "Navigating Graduate School Admissions and Challenges of Academia," *KSEA Ygnite*, San Jose, CA, US, January 2023.

T7. **J. Lee**, "Dynamical Systems Approach on Network Systems," *SNU Aerospace Seminar*, Seoul National University, Seoul, Republic of Korea, December 2022.

T8. **J. Lee**, "Dynamical Systems Approach on Network Systems," *CNU Aerospace BK21 & Colloquium Series*, Chung-Nam University, Daejeon, Republic of Korea, December 2022.

T9. **J. Lee**, "Dynamical Network Systems: Consensus Problem," *Smart Air Mobility Seminar*, Korea Aerospace University, Online, July 2022.

Professional Services

Professional Societies:

Member, Institute of Electrical and Electronics Engineers (IEEE)
- Member, IEEE Control Systems Society (CSS)
- Member, IEEE Systems, Man, and Cybernetics Society (SMCS)

Member, American Institute of Aeronautics and Astronautics (AIAA)

Member, Society for Industrial and Applied Mathematics (SIAM)
- Member, SIAM Activity Groups on Control & Systems Theory (SIAG CST)
- Member, SIAM Activity Groups on Dynamical Systems (SIAG DS)

Member, The American Society of Mechanical Engineers (ASME)

Member, Korean-American Scientists and Engineers Association (KSEA)

Reviewers:

Archival Journals
- Automatica
- IEEE Transactions on Automatic Control (IEEE TAC)
- IEEE Transactions on Systems, Man and Cybernetics: Systems (IEEE TSMC: Systems)
- IEEE Control Systems Letters (IEEE L-CSS)
- IEEE Transactions on Control of Network Systems (IEEE TCNS)
- International Journal of Robust and Nonlinear Control
- International Journal of Control (IJC)
- European Journal of Control (EJC)
- SIAM Journal on Control and Optimization (SICON)
- Nonlinear Analysis: Hybrid Systems (NAHS)
- Systems & Control Letters (SCL)
- Journal of Spacecraft and Rockets (JSR)
- Philosophical Transactions of The Royal Society A
- Applied Mathematics and Computation (AMC)
- The Journal of the Franklin Institute (JFI)
- Asian Journal of Control (AJC)
- IEEE/CAA Journal of Automatica Sinica (JAS)
- Aerospace Science and Technology (AESCTE)
- Mathematics

Research grants
- NSF Electrical, Communications and Cyber Systems (ECCS)
- NSF Foundational Research in Robotics (FRR)
- NSF Dynamics, Control and Systems Diagnostics (DCSD)

Conferences
- IEEE American Control Conference (ACC)
- IEEE Conference on Control Technology and Applications (CCTA)

- IEEE Conference on Decision and Control (CDC)
- AIAA SciTech

Conference organization:

- Associate Editor, 2024 International Conference on Unmanned Aircraft Systems (ICUAS)
- Session Chair, 2023 IEEE American Control Conference (ACC)
- Associate Editor, 2023 International Conference on Unmanned Aircraft Systems (ICUAS)
- Session Co-Chair, 2021 IEEE Conference on Control Technology and Applications (CCTA)

Teaching Experience

Instructor, University of South Carolina
AESP420: Flight & Orbital Mechanics
AESP466: Flight Dynamics & Control
EMCH367: Controls
EMCH792: Nonlinear Control
ENCP101: Introduction to Engineering (Honors, co-instructor)

Teaching Assistant, Georgia Institute of Technology
Dynamics and Control Laboratory Fall 2018, Spring 2019, Fall 2019

Teaching Assistant, Seoul National University
Theory of Flight Vehicle Guidance and Control Spring 2017, Fall 2017

Mentoring

Graduate Students
Dario Handrick Ph.D. candidate, 2024 – present
Tethered unmanned aerial vehicle systems
Shruti Jadhav M.S., Mechanical Engineering, Dec. 2025
Dual-process decision-making for long-term autonomous missions

Current Undergraduate Researchers
Dan Walters 2026 – present
Long-term UAV autonomy and battery management
Ashton Keller 2026 – present
Battery degradation in long-term UAV missions
Jack Sullivan 2026 – present
Bench setup for retractable-tether multirotor UAV testing
Jeremy Glasgow 2026 – present
Digital fixed-time control for a robotic arm
Ryan Tiburzi 2026 – present
Digital twin of the electric grid against geomagnetically induced current
Elisheba Lee 2025 – present
Thermodynamic particle swarm optimization for autonomous search
Peter Muir 2025 – present
Distributed particle swarm optimization for 3D aerial swarms
Bode DiMarzio 2025 – present
Thermodynamic particle swarm optimization for ground swarms in confined

spaces
Andersen Spina 2025 – present
Flywheel-based, human-mounted fixed-wing UAV launcher
Shashwat Yadav 2024 – present
Robust control of retractable-tether multirotor UAVs

Undergraduate Alumni
Kavan Bojappa B.S., Mechanical Engineering, May 2026
Particle swarm optimization for multi-agent rendezvous
Evan Sill B.S., Mechanical Engineering, May 2026
Thermodynamic particle swarm optimization for swarm search
Hanna Lutz B.S., Aerospace Engineering, May 2026
Digital finite-time control via implicit discretization
Tyler Hilbert B.S., Aerospace Engineering, May 2026
Dynamics and control
Mattie Eckenrode 2024 – 2025
Collaborative tethered unmanned aerial vehicles

Student Honors
- D. Walters, McNair Research Fellowship and Journey Grant for Early Researchers Award, 2026
- S. Yadav, NASA South Carolina Space Grant Consortium Undergraduate Research Award, 2026
- A. Spina, First Place, Discover USC 2026, Engineering and Computing C session, 2026
- A. Keller, Capstone Scholars Program Award, 2026
- E. Lee, Undergraduate Research Mini-Grant, 2026
- P. Muir and B. DiMarzio, Undergraduate Research Mini-Grants, 2025
- S. Yadav, Second Prize, Undergraduate Expo, ASME IMECE 2025
- A. Spina, Magellan Scholar Research Grant, 2025
- K. Bojappa, Magellan Scholar Award and Magellan Ambassador, 2024
- K. Bojappa, NASA South Carolina Space Grant Consortium Undergraduate Student Research Award and Magellan Mini-Grant, 2023